Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/30/2007
PI Name:	Sukharev, Sergei Ph.D.		
Project Title:	Mechanosensitive Channels In Bacteria: The Role Of Protein-Lipid Interactions In The Mechanism Of Gating		
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PI Email:	sukharev@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-6923
Organization Name:	University of Maryland		
PI Address 1:	Biology Department		
PI Address 2:			
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742	**Congressional District:**	5
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Task Description:	We are currently approaching only a minimal level of understanding of how cells sense and respond to mechanical stimuli. There still remains a drastic mismatch between the amount of 'functional' phenomenology data, experimental observations collected at the organismal and cellular levels, and corresponding molecular descriptions of the underlying mechanosensory mechanisms. MscL and MscS are bacterial channels which respond to membrane stretch by opening large aqueous pores. They reside in the inner membrane of Escherichia coli and serve as protective safety valves in the event of osmotic shock. Solved crystal structures tremendously facilitate progress toward elucidating the molecular mechanisms of these tension-driven valves. They also aid in the search for a general understanding of the biophysical principles that govern these primary events in mechanosensation. During the previous funding periods, we determined the overall structure of MscL, and evaluated the energetic and spatial parameters that characterize its conformational transitions. We have built a plausible molecular model of MscL conformational transition. Since MscS was identified		

	and crystallized later, there is currently less data on its mechanisms and the models of its gating are less precise. Our methodology includes site-directed mutagenesis, covalent protein modifications, patch-clamp measurements, thermodynamic and kinetic analysis of dose-response curves and transition rates, video-microscopy, molecular modeling, and simulations. The goal is a consistent biophysical description of gating of these two structurally dissimilar mechanosensitive channels at an atomic-scale level.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research leads to understanding of very basic mechanisms of force detection by macromolecules in living organisms. It will help to clarify general principles and mechanisms of mechanosensation, which encompass a wide range of phenomena from simple osmoregulation in bacteria to complex phenomena such as gravitropism in plants and balance and hearing in humans. MscL represents simple and powerful model system highly useful for molecular and biophysical experiments as well as for molecular computation. Obtaining a complete and self-consistent picture of mechanosensitive channel gating which includes the protein, surrounding lipids and solvent would be a stepping stone to understanding the mechanisms of the growing families of sensory channels modulated by membrane stretch cell volume, lipids, amphipathic substances, or temperature.
Task Progress:	During the past funding period we had been able to address several important questions related to the role of protein-lipid interactions in the gating of bacterial mechanosensitive channels. These channels are embedded in the lipid bilayer and the mechanical stimulus is transmitted to their gates through the lipid-protein interface. The first question was the roles of tyrosines and other aromatic residues at the rims of the large mechanosensitive channel MscL as anchors for the gating and the osmotic rescuing function in vivo. The results strongly suggested that the absence of aromatic anchors on the cytoplasmic rim of the channel barrel permits for a lesser energetic penalty of lipid distortion at the protein-lipid interface (thickness mismatch) during the opening process associated with a substantial flattening of the barrel. The second question that has been explored was the thermal dependence of MscL gating. In experiments, MscL was found to be cold-activated. Searching for the molecular players and groups responsible for cold sensitivity we performed molecular dynamic simulations of the open and closed MscL in fully hydrated lipid bilayer. The results revealed the character of lipid-protein interactions and lipid distortion near the boundary but suggested that lipids are the unlikely candidates for the role of temperature sensor in this system. We have completed energetic characterization of MscL gating and provided compelling evidence for the two-gate mechanism of this channel. Three papers have been published in the leading journals in the field and two more papers are in preparation. See previous year's reports for all journal articles published as a result of NASA support.
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